[Abstract]

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract: We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers.

As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

[Masked Language Modeling & Next Sentence Prediction]

3.1 Pre-training BERT: Building on the Transformer encoder architecture proposed by Vaswani et al. (2017), we pre-train BERT using two unsupervised tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In Masked LM, we randomly mask 15% of all WordPiece tokens in each sequence and predict the masked tokens.

For the 15% of tokens selected for masking, we replace the token with the [MASK] token 80% of the time, a random token 10% of the time, and the unchanged token 10% of the time. Next Sentence Prediction trains a binary classifier to determine whether sentence B actually follows sentence A in the pre-training corpus.

[GLUE & SQuAD Benchmark Results]

4 Experiments: BERT obtains new state-of-the-art results on eleven natural language processing tasks. It pushes the GLUE benchmark score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement), and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).